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Identification of critical nodes in multimodal transportation network

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ABSTRACT

A transportation network is an essential lifeline engineering system, and its reliability is critical when faced with natural or man-made disasters. The reliability of the transportation network will affect the decision-making process of the managers of a country or province during disasters. When a disaster strikes, one or more critical nodes in the transportation network may completely lose their basic function, which may greatly reduce the reliability of the transportation network. Therefore, identifying critical nodes in the transportation network is of utmost importance in the analysis of the reliability of the transportation network. The complex network theory provides a powerful tool to identify critical nodes. In this study, we propose an improved weighted k-shell (IWKS) model to identify the critical nodes based on the complex networks theory. This model comprehensively considers the diversity of the transportation modes, independent transportation ability, and connectivity of the node. Additionally, the comprehensive transportation network in Zhejiang (China) was used to illustrate the effectiveness of the proposed method. The results show that the proposed method can effectively identify critical nodes in the multimodal transportation network (MTN). We considered three indicators in this model, and it is also feasible to select a few more indicators. This study may provide a new perspective for the application of the complex network theory in the transportation system, and may also provide the necessary information to improve the reliability of the network.

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1. Introduction

The transportation system is an essential component of the robust operation between cities. In the face of natural or man-made disasters, the reliability of the transportation network is critical as they provide crucial access to affected areas. Reliability refers to the probability that the transportation network remains functional under disrupting conditions [1]. The differences in the reliability of the transportation network will lead national or provincial managers to make different decisions when a disaster strikes. However, knowledge of the complexity and topological structure of the transportation network is critical for a better understanding of its reliability during a disaster.

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The research of complex networks [2] has blossomed in the last decade [3]. Many real-world systems, such as transportation systems, social systems, biological systems, and communication systems, can be described through the complex network [4]. It provides an important insight for understanding the transportation network structure, function, and complexity [3,5–7]. It is a common way to abstract the transportation network into a simple network composed of nodes and edges based on the complex network theory [8,9]. In the past, extensive research has been conducted on unimodal transportation networks, such as railway network [10–12], air transport network [13,14], waterway network [15,16], bus network [17], subway network [18] or road network [19], with the stations and connecting links as nodes and edges, respectively. However, these networks are not independent, and the coupling between networks will change the characteristics of the system [20]. As a result, the multimodal transportation network (MTN) theory [21–23] has gradually developed in the last few years [24]. The MTN is modeled as a multilayer network with different layers corresponding to different transportation modes [22], such as the coupled of railway and air transport network [25], road and railway network [26], bus and subway network [23,27], and bus, tram, and subway network [21]. The nodes of an MTN are generally abstracted by transportation hubs, counties, or cities. This is like abstracting the network at the micro, meso, or macro level. With different granularities, the nodes play different roles in the network. At the micro-level, the failure of a transportation hub has little impact on intercity traffic. At the macro-level, the size of the city is too large, and the probability of failure of the entire transportation hub of the city is small. Therefore, it is appropriate to abstract the districts and counties as nodes at the meso-level. Critical nodes [28] are a set of nodes that have a significant impact on the network structure, function, and reliability. The failure of a small fraction of critical nodes in the transportation network will greatly reduce transportation efficiency and network reliability, which will delay the emergency evacuation of people and the arrival of rescuers. Therefore, the critical nodes in the transportation network are critical to guarantee the normal operation of the network. Thus, how to find critical nodes has theoretical significance and practical value for the analysis of the transportation network reliability.

There are various approaches to determine the node importance. Previous studies commonly used a single centrality to rank nodes, such as degree centrality, betweenness centrality, closeness centrality, eigenvector centrality, k-shell, *etc*. The definitions of these measures were summarized by Lü et al. [28] Recently, a new trend has emerged to comprehensively identify critical nodes by combining multiple indicators [29–31]. Among them, some methods considering both local and global metrics [32,33] combined the advantages of the lower computational complexity of local metrics with higher computational accuracy of global metrics. Therefore, constructing a hybrid measure is useful to reasonably evaluate the importance of the node. However, it is not possible to find a universal measure that best quantifies the importance of the node in every scenario [28]. Thus, we need to develop a critical node identification method suitable for transportation networks based on existing methods and transportation network characteristics and functions. In general, the application of complex networks in transportation systems still needs further exploration.

Motivated by these factors, we propose a novel method to identify the critical nodes of MTN based on complex networks. To describe the comprehensiveness and diversity of the transportation modes, we adopt an information entropy model to construct a multimodal composite transportation network (MCTN), which is a composite of the highway network, conventional railway network, high-speed railway network, aviation network, and waterway network. Then, to identify the critical nodes, we propose an improved weighted k-shell (IWKS) model, which comprehensively takes into account the diversity of transportation modes, independent transportation ability, and connectivity of the node. Ultimately, the comprehensive transportation network of Zhejiang was used to illustrate the effectiveness of the proposed approach.

The remaining parts of this paper are organized as follows. Section 2 introduces the methodology in detail. Experiment results are analyzed in Section 3. Lastly, the conclusions of this study are presented in the final section.

2. Methodology

There are basic differences between passenger and freight transport due to the difference between their service objects. Accordingly, their network should be separately constructed to identify critical nodes. Although, it is believed that their ways of constructing the network topology model and identifying critical nodes are similar. Therefore, the passenger transportation network is used as an example to illustrate the methodology.

2.1. Network modeling

2.1.1. Topology model of a single-layer transportation network

Four main modes of transportation are considered to build the topology network: highway, railway, aviation, and waterway. To more accurately analyze the characteristics, the railway is divided into conventional railway and high-speed railway due to the large differences in operating speed, operating time, and transportation capacity. Accordingly, five single-layer transportation networks are constructed, namely the highway network G_h, conventional railway network G_r, high-speed railway network G_s, aviation network G_f, and waterway network G_w.

To analyze the structure of the transportation network from the perspective of topological properties, the space L [34–36] is used to construct an undirected weighted transportation network. The transportation network is abstracted as a graph $G = (V, E, W)$ consisting of an undirected graph $G = (V, E)$ with a node set V, an edge set E, and an initial weight

Table 1
Node types of MCTN.

Node types	Node meaning
A-type	Single station, single transportation mode, single service object
B-type	Multiple stations, single transportation mode, single service object
C-type	Single station, multiple transportation modes, multiple service objects
D-type	Multiple stations, multiple transportation modes, single service object
E-type	Multiple stations, multiple transportation modes, single + multiple service objects

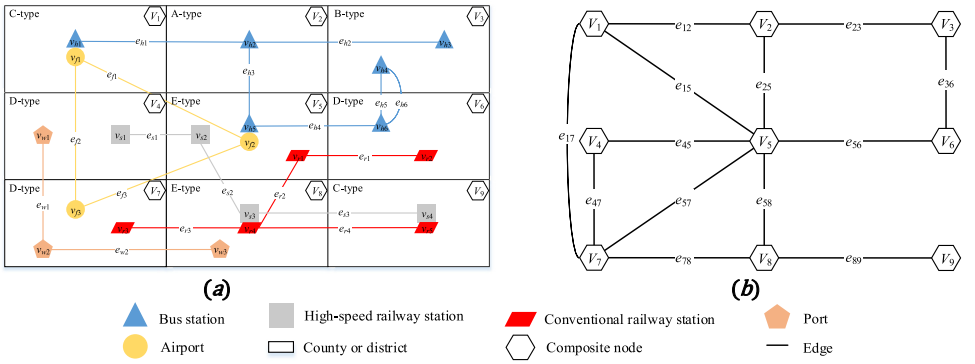


Fig. 1. An example of a multimodal transportation network. $\boldsymbol{V_1}$-$\boldsymbol{V_9}$ represents 9 counties, namely, the composite nodes of the MTCN. Different colors indicate different transportation modes. If two stations are next to each other, it means they belong to the same transportation hub, that is, passengers have two transportation modes to choose from at this transportation hub. (a) The topology of five single-layer transportation networks, and (b) The topology of MTCN.

function $W = (t, d)$, where operating time t and distance d are the initial edge attributes of each edge. Because different transportation modes have different effects on the MCTN, and MCTN will show new characteristics, it is necessary to add new edge attributes and calculate the final weight function of the MCTN, which will be described in the next section. Besides, $N = |V|$ is the number of nodes. The space L defines the bus stations, conventional railway stations, high-speed railway stations, airports, or ports as nodes. If two stations are adjacent in at least one trip, an edge connects the two nodes.

2.1.2. Topology model of the MCTN

To build the topology model of the MCTN G_c based on five sub-transportation networks, two principles are followed: (1) Consider the county or district (hereafter referred to as "county") where the bus station, conventional railway station, high-speed railway station, airport, or port is located as a composite node. If the two counties are adjacent in at least one trip, an edge connects the two counties; (2) If there are multiple transportation modes from county S to county T, it is considered that there is only one composite edge connecting the two counties in the MCTN. The composite edge refers to the form of abstracting multiple edges of the single-layer transportation network into one edge of the MCTN.

To make the abstracted MCTN reflect the properties, functions, and characteristics of the real transportation network, the composite nodes of the MCTN are divided into five types according to the transportation mode, service object, and number of stations, as shown in Table 1, namely: (1) The A-type composite node indicates that there is only one station in a county and its service object is unique, that is, passengers have only one mode of transportation to choose from at this station. For example, there is only one bus station in the county. (2) The B-type composite node means that there are two or more stations in a county. These stations belong to the same transportation mode, and the service object is unique. For example, there are two bus stations in the county. (3) The C-type composite node refers to when there is one station in a county, which includes different transportation modes and its service objects are different, *i.e.*, there are two or more modes of transportation for passengers to choose from at this station. Some transportation hubs serve multiple transportation modes, for instance, some railway stations serve both conventional railway and high-speed railway or most airports have an affiliated bus station that allows passengers to take long-distance buses without leaving the airport. (4) The D-type composite node indicates that there are multiple stations in a county. Each station corresponds to a transportation mode and serves a unique service object. For example, a county has a bus station and a high-speed rail station. (5) The E-type composite node means that there are multiple stations in a county. Some stations only provide one mode of transportation and serve a unique object, while other stations provide multiple transportation modes and serve different objects. For example, a county has a bus station and a railway station, where the railway station serves both conventional railway and high-speed railway. To provide an intuitive explanation of the composite node type, we present the example of a simple multimodal transportation network with 9 composite nodes in Fig. 1.

Then the composite edge in the MCTN needs to be weighted. Since the five types of composite nodes classified in MCTN are characteristics not found in single-layer networks, we add the edge type to the edge attributes of MCTN to reflect the diversity of transportation modes. That is, we select the operating time t, distance d, and edge type c as edge attributes to calculate the final weight of MCTN. The edges between different composite node types have different transportation capacities and functions, which can be illustrated by the different edge type values. The edge type value is calculated by combining the composite node type, node number, and transportation mode, which is expressed as:

$$c = \sigma_k \left(\theta_i \sum_{k=1}^{5} N_i^k \delta_i^k + \theta_j \sum_{k=1}^{5} N_j^k \delta_j^k \right) \tag{1}$$

where σ_k is the edge type coefficient, *i.e.*, the coefficient corresponding to the edge of transportation mode k; θ_i and θ_j are the type coefficient of composite nodes V_i and V_j, respectively; N_i^k and N_j^k denote the node number of transportation mode k included in composite nodes V_i and V_j, respectively; δ_i^k and δ_j^k denote the transportation mode coefficient of transportation mode k in composite nodes V_i and V_j, respectively; $k = 1, 2, 3, 4, 5$ indicates five modes of transportation.

A method based on the information entropy theory [37] is widely used to calculate the weight without any additional or subjective information [38]. The calculation process is as follows. Let $E = (e_1, e_2, \ldots, e_m)$ be the edge set between the MCTN nodes, *i.e.*, there are m edges. Let $Q = (q_1, q_2, \ldots, q_n)$ be the set of edge attributes. The weight of the edge e_x $(x = 1, 2, \ldots, m)$ is measured by the attribute q_y $(y = 1, 2, \ldots, n)$, and the obtained attribute value a_{xy} of e_x with respect to q_y forms the decision matrix $A = \left(a_{xy}\right)_{m \times n}$.

$$A = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{vmatrix} \tag{2}$$

To eliminate the influence of different physical dimensions, the decision matrix is standardized using Eq. (3) or Eq. (4) to obtain the matrix $R = \left(r_{xy}\right)_{m \times n}$.

$$\text{Cost type:} \quad r_{xy} = \min_x \left(a_{xy}\right) / a_{xy}, \ x = 1, 2, \ldots, m; \ y = 1, 2, \ldots, n \tag{3}$$

$$\text{Benefit type:} \quad r_{xy} = a_{xy} / \max_x \left(a_{xy}\right), \ x = 1, 2, \ldots, m; \ y = 1, 2, \ldots, n \tag{4}$$

The matrix R is normalized to obtain the matrix $H = \left(h_{xy}\right)_{m \times n}$, where

$$h_{xy} = r_{xy} / \sum_{x=1}^{m} r_{xy}, \ x = 1, 2, \ldots, m; \ y = 1, 2, \ldots, n \tag{5}$$

Thus, the information entropy value H_y of the attribute q_y is obtained.

$$H_y = -\frac{1}{\ln m} \sum_{x=1}^{m} \left(h_{xy} \cdot \ln h_{xy}\right), \ y = 1, 2, \ldots, n \tag{6}$$

The weight vector of the attributes is $w_y = (w_1, w_2, \ldots, w_n)$, where

$$w_y = \left(1 - H_y\right) / \sum_{y=1}^{n} \left(1 - H_y\right), \ y = 1, 2, \ldots, n \tag{7}$$

The weight w_x of the edge e_x can be expressed as

$$w_x = \sum_{y=1}^{n} w_y \cdot r_{xy}, \ x = 1, 2, \ldots, m; \ y = 1, 2, \ldots, n \tag{8}$$

If there are multiple edges in transportation mode k between composite nodes V_i and V_j, the edge weight W_{ij}^k of transportation mode k between composite nodes V_i and V_j can be expressed as

$$W_{ij}^k = \sum_x w_x \tag{9}$$

The composite edge weight W_{ij} between composite nodes V_i and V_j in MCTN is:

$$W_{ij} = \sum_{k=1}^{5} \alpha_k W_{ij}^k \tag{10}$$

where α_k is a parameter indicating whether there is a highway, conventional railway, high-speed railway, airline, or waterway between the MCTN node pairs. $\alpha_k = 1$, if there is a mode of transportation, otherwise $\alpha_k = 0$.

2.2. Critical nodes identification model

Many methods have been proposed to identify the critical nodes in complex networks. Among them, some methods identify critical nodes based on a single index, such as the degree centrality, betweenness centrality, closeness centrality, eigenvector centrality and k-shell, *etc*. There are also several methods to identify critical nodes based on multiple indicators that consider both global and local measures. These methods have their advantages and limitations. We should select the proper approach to identify critical nodes according to the characteristics of the network. Accordingly, we develop a new method for undirected weighted MCTN, which can characterize the diversity of transportation modes, independent transportation ability, and connectivity of the node.

For weighted networks, Garas et al. [39] proposed a weighted k-shell method considering both the number of neighbor nodes and edge weight.

$$s_i = \left[k_i^{\lambda} \left(\sum_{j}^{k_i} w_{ij} \right)^{\eta} \right]^{\frac{1}{\lambda+\eta}} \tag{11}$$

where s_i is the node strength, k_i is the degree, w_{ij} indicates the edge weight between the node V_i and its neighboring node V_j, λ and η are the adjustment parameters.

In the MCTN model developed above, the composite edge weight can reflect the diversity of the transportation modes. The node degree k_i is the number of edges connected to the node i in the network. The degree only considers the connection itself, *i.e.*, the more neighboring nodes a node has, the more important the node is. However, the degree can only reflect the importance of the node in the local network. In a composite transportation network, the importance of the node also needs to consider its function in the global network. The closeness centrality holds that the central node has the smallest total distance from this node to all other nodes. The counties with large closeness centrality value are in the center of the global network and easier to reach other counties. The closeness centrality can be used to measure the degree to which a county is not controlled by other counties. In the transportation network, the closeness centrality reflects the independent transportation ability of the county [40]. Therefore, the following formula is obtained by considering the closeness centrality based on Eq. (11).

$$s_i = \left[k_i^{\lambda} \left(\sum_{j}^{k_i} W_{ij} \right)^{\eta} C_c(i)^{\mu} \right]^{\frac{1}{\lambda+\eta+\mu}} \tag{12}$$

where λ, η and μ are the adjustment parameters, the meaning of λ and η are same as in Eq. (10); W_{ij} represents the composite edge weight of the MCTN between the node V_i and its neighboring node V_j; $C_c(i)$ is the closeness centrality [28], which is expressed as:

$$C_c(i) = \frac{N-1}{\sum_{j=1, j\neq i}^{N} d_{ij}} \tag{13}$$

where d_{ij} denotes the shortest distance between node V_i and V_j, N denotes the total number of nodes in the network.

Since the dimensions of each parameter are different, the parameter is standardized using Eq. (4). To avoid the subsequent calculation value being too small, this formula increases the standardized value by 10 times, and the value range is [0, 10]. We use k_i^*, $\sum_j^{k_i} W_{ij}^*$ and $C_c^*(i)$ to represent 10 times the standardized parameter value. Then, Eq. (12) becomes Eq. (14), which is the IWKS model of the MCTN proposed in this study.

$$s_i^* = \left[\left(k_i^*\right)^{\lambda} \left(\sum_{j}^{k_i} W_{ij}^* \right)^{\eta} C_c^*(i)^{\mu} \right]^{\frac{1}{\lambda+\eta+\mu}} \tag{14}$$

where s_i^* is the node importance.

3. Experiments and results

To evaluate the effectiveness of the IWKS model, we apply it to the real transportation network of Zhejiang, China. The overall construction level of the transportation infrastructure in Zhejiang Province is high, and it has a certain representation.

Table 2
The coefficients of Eq. (1).

	A-type	B-type	C-type	D-type	E-type
θ_i, θ_j	1.2	2	3	4	5
	Waterway	Aviation	Highway	Conventional railway	High-speed railway
σ_k, δ_i^k, δ_j^k	1.2	1.5	2	2.5	3



Fig. 2. The actual locations and codes of the districts and counties in Zhejiang.

3.1. The characteristics of the comprehensive transportation network in Zhejiang

The data of the operating time t and distance d of each transportation mode are obtained from ticketing websites and map websites. The composite edge type value c is calculated by Eq. (1). Since each composite node type contains different transportation modes, the number of stations and service objects, the composite node type coefficients θ_i, θ_j of types A, B, C, D, and E are assigned values of 1.2, 2, 3, 4, and 5, respectively. Besides, both the edge type coefficient σ_k and transportation mode coefficients δ_i^k, δ_j^k are related to the transportation mode, therefore, these three coefficients take the same range. Different values were taken according to the transportation capacity, transportation efficiency, and service level of each transportation mode respectively, as shown in Table 2. The actual locations and codes of the districts and counties (*i.e.*, composite nodes) in Zhejiang are shown in Fig. 2. Although there are no intra-provincial airlines opened in Zhejiang, in the extraordinary period of disaster, we assume that there are direct routes between airports, *i.e.*, airports are interconnected. Then the MCTN topology is constructed by the modeling method proposed in Section 2.1, with a total of 89 composite nodes and 268 composite edges, as shown in Fig. 3.

Based on the Python NetworkX 2.2 framework, we simulated and calculated the complex statistical characteristics of the MCTN, and each transportation mode network in Zhejiang is shown in Table 3. This data revealed that the average degree and maximum degree of the MCTN are greater than those of the single-mode transportation network. In other words, the composite network provides more transportation modes and transportation route options for passenger transportation. It also improves traffic accessibility and network connectivity. The average shortest path length represents the average number of nodes that the shortest path between any two nodes in the network needs to pass. For the comprehensive transportation network, this indicator reflects the ease of access between the two counties. Except for the special aviation network, the average shortest path length of the composite network is less than that of the single-mode transportation network, which indicates that the composite network improves the accessibility between the two counties. The larger the average clustering coefficient, the closer the connection between nodes and the better the network connectivity. Except for the special aviation network, the average clustering coefficient of the composite network is higher than that of the single-mode transportation network, which indicates that the composite network has more obvious communalization, higher aggregation, and better connectivity and accessibility. In short, these findings show that the composite network has the complementary advantages of various transportation modes.

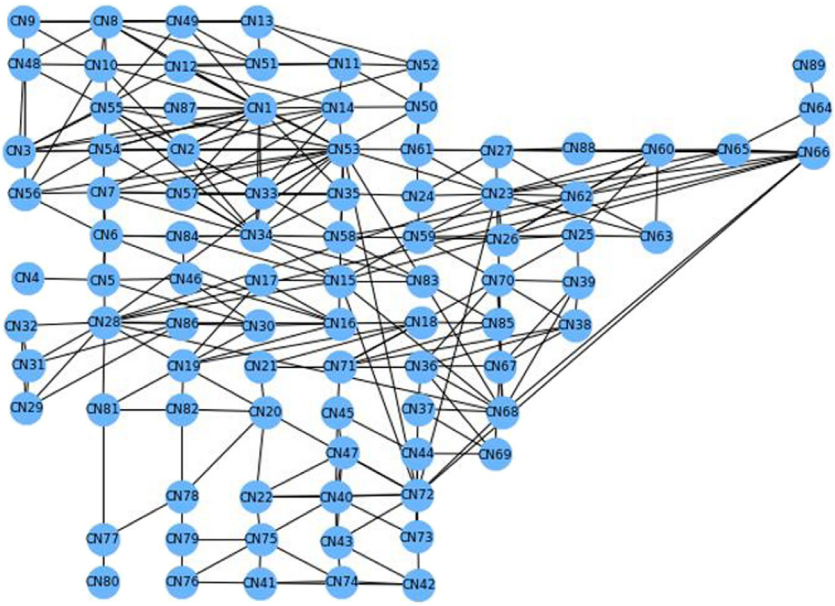


Fig. 3. MCTN topology of Zhejiang.

Table 3
The statistical characteristics of network topology.

	Highway network G_h	Conventional railway network G_r	High-speed railway network G_s	Aviation network G_f	Waterway network G_w	MCTN G_c
Number of nodes	106	27	47	7	62	89
Number of edges	272	31	50	21	56	268
Average degree	5.1321	2.2963	2.1277	6	1.8065	6.0224
Maximum degree	16	4	6	6	3	19
Average clustering coefficient	0.4045	0.0679	0.0475	1	0	0.4142
Average shortest path length	5.0429	5.1140	7.6503	1	8.4571	3.2901

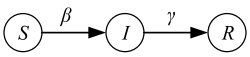


Fig. 4. Spread mechanism of the SIR model.

3.2. Susceptible–Infected–Recovered model

Similar to other studies [2,41], we adopt the susceptible–infected–recovered (SIR) model to evaluate the accuracy of the IWKS. The SIR model assumes that the nodes have three states: (1) susceptible $S(t)$ indicates the proportion of individuals susceptible to (not yet infected) the disease in the total number of nodes at time t; (2) infected $I(t)$ represents the proportion of individuals who have been infected and can further spread the disease to susceptible individuals in the total number of nodes at time t; (3) recovered $R(t)$ denotes the proportion of individuals who have been recovered and will never be infected again in the total number of nodes at time t. The Spread mechanism of the SIR model is shown in Fig. 4. In the transportation network, the station where the emergency occurred can be regarded as an infected node and the others can be regarded as susceptible nodes. Then the infected node selects susceptible neighbors to get infected with a probability β. Infected nodes will recover with a probability γ after some time. The recovered nodes will always be in this state.

The general SIR model is used to evaluate the unweighted network, which is different from the weighted network constructed in this study. The weight of an edge represents the close relationship between two connected nodes, which should be considered in the propagation process of the impact of an emergency. Therefore, the infected probability between composite nodes V_i and V_j is modified to $\beta_{ij} = W_{ij}/W_{\max}$, where $W_{\max}$ is the maximum value of all W_{ij} [42]. However, the infected node becomes an immune node that is usually not affected by the weight of the edge. Thus, the recovery probability remains unchanged, let $\gamma = 1$. The spreading influence refers to all nodes that have been infected by the spread of the disease. To investigate the spreading influence of a node, we set it as the initially infected node. The spreading influence of the initially infected node at time t is the total number of infected and recovered nodes at time t, denoted by $F(t)$, that is, $F(t) = I(t)+R(t)$ [43].

Table 4
Comparison of correlation coefficients corresponding to different adjustment parameters.

λ	η	μ	Kendall's tau	Spearman	Pearson	λ	η	μ	Kendall's tau	Spearman	Pearson
1	1	1	0.6743	0.8384	0.8831	1	2	3	0.6629	0.8245	0.8464
1	1	2	0.6832	0.8490	0.8819	2	1	3	0.7305	0.8856	0.8457
1	2	1	0.5744	0.7501	0.7967	1	3	2	0.5861	0.7619	0.7439
2	1	1	0.7023	0.8754	0.8754	2	3	1	0.5987	0.7709	0.6802
1	2	2	0.6307	0.8069	0.8427	3	2	1	0.6869	0.8489	0.7709
2	1	2	0.7598	0.9108	0.8821	3	1	2	0.7419	0.9038	0.8058
2	2	1	0.6616	0.8262	0.8280						

3.3. IWKS model for the comprehensive transportation network in Zhejiang

First, we determined the adjustable parameters λ, η and μ of the IWKS model. The adjustable parameters λ, η and μ can be taken as values from 1, 2, or 3, which represent the different importance of the corresponding indicators (degree, composite edge weight, and closeness centrality). We believed that the two types of values $\lambda = 1$, $\eta = \mu = 2$ and $\lambda = 1$, $\eta = \mu = 3$ represent the same meaning, *i.e.*, the composite edge weight and closeness centrality are equally important and more important than degree, so we chose only one of them for the experiment. Thus, we selected 13 different parameter combinations for experimental comparison. To determine the adjustable parameters, the SIR model is used to measure the real spreading influence of the composite nodes. We assume that when some counties (initially infected composite nodes) experience a disaster that causes a failure, the adjacent counties (susceptible composite nodes) will fail with a probability β_{ij}. In general, the average number of infected nodes increases as the transmission time increases, and eventually reaches a stable value [44]. And we use Kendall's [45], Spearman's [46], and Pearson's [47] correlation coefficients between IWKS and the real spreading influence in the 5-time steps (F(5)) to evaluate the accuracy of IWKS under different adjustable parameters. The higher the value of the correlation coefficient, the more accurate the model. The correlation coefficients corresponding to different adjustment parameters are listed in Table 4.

From Table 4, it can be seen that when $\lambda = \mu = 2$, $\eta = 1$, the overall performance of the three correlation coefficients is the best, *i.e.*, the model is the most accurate. Then the IWKS model of the comprehensive transportation network in Zhejiang can be obtained, which is expressed as:

$$s_i^* = \sqrt[5]{\left(k_i^*\right)^2 \left(\sum_j^{k_i} W_{ij}^*\right) C_c^*(i)^2} \tag{15}$$

3.4. Analysis of critical nodes

To evaluate the accuracy of the method, some well-established methods, such as degree, betweenness centrality [48], closeness centrality, k-shell, and structural holes [49] are selected as benchmark methods for performance evaluation. The analysis of critical nodes is divided into five parts, as described below.

3.4.1. Comparison of the real spreading influence of the composite nodes

The top-10 ranked nodes by degree, betweenness, closeness, k-shell, structural holes, SIR, and IWKS are obtained by averaging over 100 implementations and are listed in Table 5. The results of the closeness centrality show that the top-7 composite nodes (CN53, CN28, CN72, CN23, CN15, CN68, and CN66) all include the airport. The reason for this is that the haul distance between airports is the shortest of all modes of transportation in long-distance passenger transport, and it shortens the distance to other nodes, thereby resulting in a larger value of closeness centrality. This indicates that these nodes have strong independent transportation ability and are in the center of the global network. Meanwhile, the betweenness centrality and k-shell of these nodes are also ranked in the top-7, which indicates that they have the strongest transfer capability and are in the center of the network. In addition, the degree value of these nodes ranked in the top-11, indicating that there are more adjacent nodes. That is to say, these nodes ranked high in both local network and global network. This is consistent with the calculation result of the IWKS method because these nodes are ranked in the top-10. The other three top-10 nodes calculated by the IWKS method are CN1, CN34, and CN33 (*i.e.*, Jianggan, Zhuji, and Yuecheng), which are important transportation hubs or located in important locations along the transportation corridor. The above analysis shows that the calculation results of the IWKS method are reasonable both at the physical and transportation levels.

In practical applications, we pay more attention to the top-ranked node set rather than one node in particular. Therefore, the top-10 nodes can be divided into 3 levels, namely the top-3, top-5, and top-10. The top-3, top-5, and top-10 composite nodes of each method are selected as initially infected origins, and the results are averaged over 100 independent experiments, as shown in Fig. 5. These results reveal that the real spreading influence of the IWKS method is higher than that of the other methods, which indicates that IWKS outperforms the other methods. We also find that there is no much difference between the final influence of the top-5 nodes and top-10 nodes, since after a certain time step, half of the nodes will be a failure. Therefore, the top-5 nodes should be regarded as key protection areas.

Table 5
Comparison of critical nodes identification results.

Rank	Degree	Betweenness	Closeness	K-shell	Structure holes	SIR	IWKS
1	CN53	CN53	CN53	CN53	CN53	CN53	CN53
2	CN1	CN72	CN28	CN68	CN23	CN34	CN23
3	CN23	CN28	CN72	CN28	CN1	CN23	CN28
4	CN34	CN68	CN23	CN15	CN34	CN1	CN1
5	CN28	CN23	CN15	CN23	CN58	CN15	CN15
6	CN72	CN15	CN68	CN72	CN25	CN14	CN72
7	CN15	CN66	CN66	CN66	CN15	CN33	CN34
8	CN55	CN81	CN34	CN34	CN59	CN28	CN68
9	CN33	CN16	CN54	CN33	CN28	CN68	CN66
10	CN68	CN1	CN1	CN54	CN72	CN66	CN33

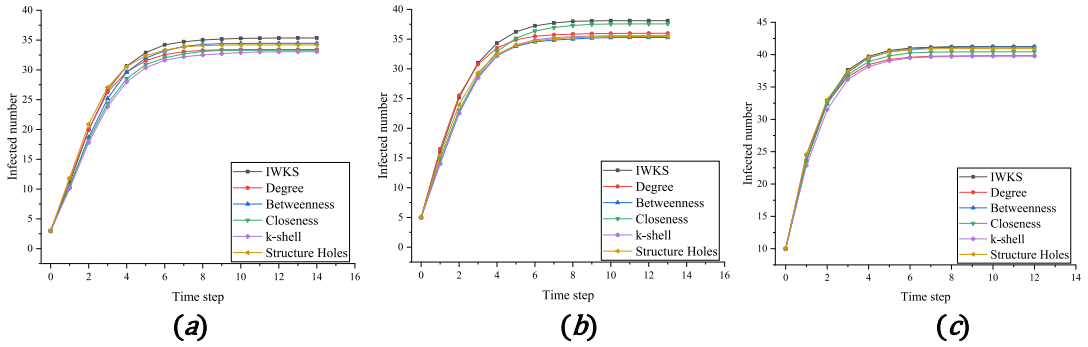


Fig. 5. The real spreading influence of each method under different proportions of initially infected nodes. (a) The top-3 nodes, (b) The top-5 nodes, and (c) The top-10 nodes are set as initially infected nodes.

Table 6
The number of each node type among the top-5 nodes.

Node type	Degree	Betweenness	Closeness	K-shell	Structure holes	IWKS
A	0	0	0	0	1	0
B	0	0	0	0	0	0
C	0	0	0	0	0	0
D	1	3	2	2	1	1
E	4	2	3	3	3	4

Table 7
The number of each node type among the top-10 nodes.

Node type	Degree	Betweenness	Closeness	K-shell	Structure holes	IWKS
A	0	1	0	0	2	0
B	0	0	0	0	0	0
C	0	0	0	0	0	0
D	5	4	5	6	3	5
E	5	5	5	4	5	5

3.4.2. Comparison of the number of each node type between the IWKS and conventional methods

The number of each node type in each level presented in Tables 6 and 7 reveal that the E-type and D-type nodes account for a larger proportion of the nodes, and other types of nodes are almost zero. Previous results have confirmed that the most important node ranked by the IWKS method is also the node with the highest degree, closeness, and k-shell. In other words, the E-type and D-type nodes with a large degree, closeness centrality, and k-shell value are critical nodes. Therefore, we should focus on protecting them to improve the reliability of the network against risks. These nodes have multiple transportation modes, large traffic capacity, strong transfer capacity, and many functions. Additionally, most of them are administrative and economic centers, occupying an important position in the province. However, the A-type, B-type, and C-type nodes are generally located at the border of the network or the end of the path. Such nodes have a single function and few transportation and transfer requirements. Therefore, these findings suggest that it is necessary to divide the composite nodes into multiple types, which can more clearly distinguish the importance of the nodes.

Table 8
The node coincidence rate of each method (%).

Rank	Degree	Betweenness	Closeness	K-shell	Structure holes	IWKS
Top-5	80	40	60	60	80	80
Top-10	80	70	80	80	60	90

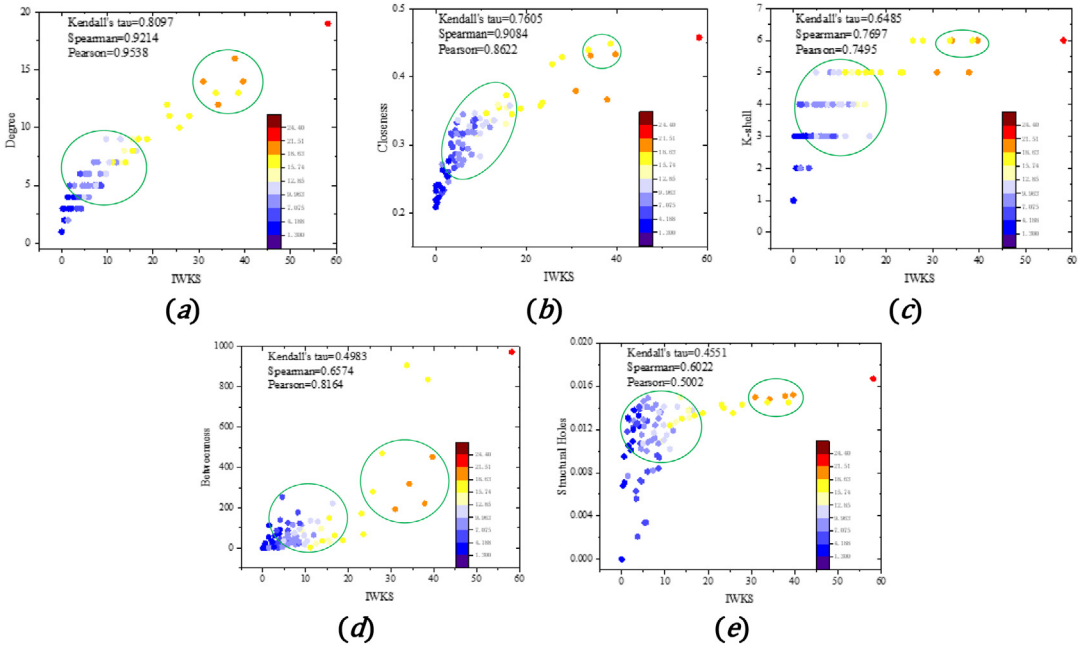


Fig. 6. The correlations between IWKS and degree, closeness, k-shell, betweenness, and structure holes.

3.4.3. Comparison of the node coincidence rate between the IWKS and conventional methods

Here, we define the node coincidence rate as the ratio of the same node number between the result of a method and the result of the SIR model to the total number of nodes in each level. The node coincidence rate of each method shown in Table 8 immediately reveals that the node coincidence rate of the IWKS method is the highest at all levels, which indicates that the overall performance of the IWKS method is better than that of the other methods. Thus, it can be concluded that from this perspective the proposed method is more effective than the others to identify critical nodes.

3.4.4. Comparison of the relation between different methods

The correlations between the IWKS method and other methods are displayed in Fig. 6. Each point represents a node in the network, and the corresponding color indicates the influence value of this node in the 5-time steps. The values are obtained by averaging over 100 independent experiments. The results show that the IWKS method has the strongest correlation with the degree and closeness centrality. However, the colors in the green areas of Fig. 6(a) and (b) are rather chaotic, indicating that the nodes with large values calculated by a single index such as degree and closeness centrality may not have a high real spreading influence. This is also reflected in the graph of the k-shell, betweenness centrality, and structural holes, which means that the traditional single-index method has shortcomings in identifying critical nodes of the comprehensive transportation network. Therefore, it is reasonable for the IWKS method to comprehensively consider the advantages of degree centrality, closeness centrality, and k-shell to identify critical nodes.

3.4.5. Comparison of the relation between each method and real spreading influence

The relations between the different methods and real spreading influence in the 5-time steps (F(5)) presented in Fig. 7 show that IWKS identified critical nodes with much higher accuracy than closeness, structure holes, and betweenness according to the correlation coefficient value. And compared with the degree and k-shell, which have similar identification accuracy, IWKS has a greater advantage in the identification of node importance distinction. Since there are many nodes with the same degree value, it is not possible to evaluate the importance of nodes by degree only. In particular, the transportation network is similar to a power-law distribution, in which most nodes have small and equal degree values so that it is difficult to distinguish the importance of the nodes. It can also be seen from Fig. 7(b) that some nodes with

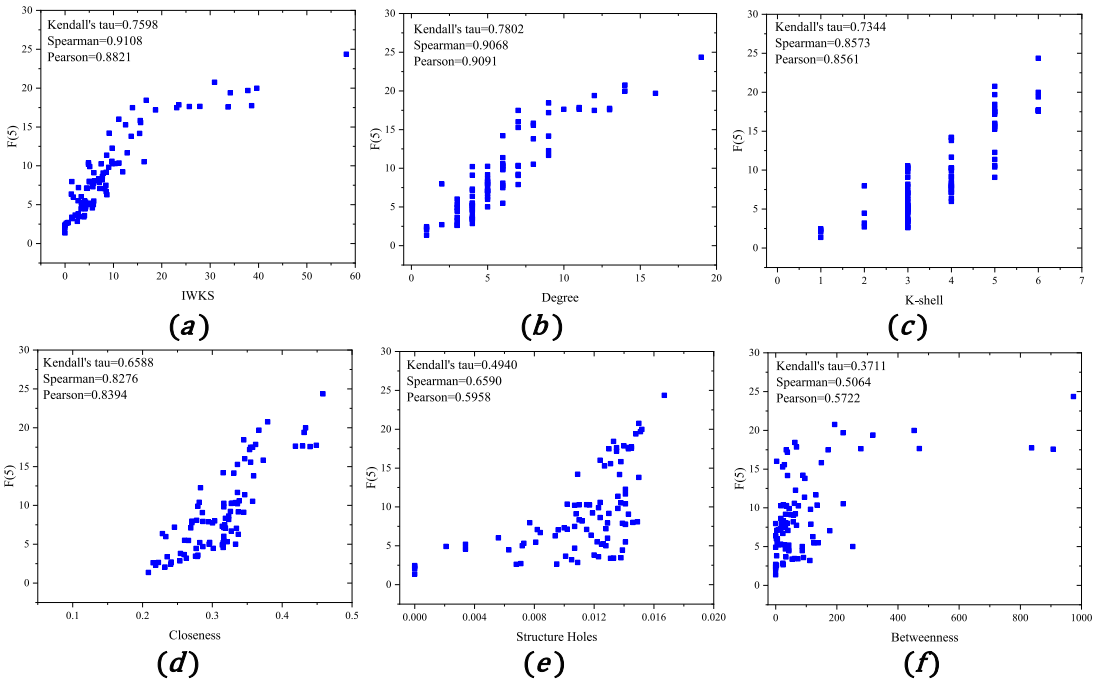


Fig. 7. The correlations between each method and real spreading influence.

higher degree values also have the same degree value. Therefore, it is not effective to distinguish the importance of nodes only by degree value. In addition, k-shell has the same shortcomings as the degree. Although the proposed method is based on k-shell, it compensates for the shortcoming of low discrimination by comprehensively taking into account the diversity of transportation modes (weight), independent transportation ability (closeness centrality), and connectivity of the node (degree). The above results indicate that the overall performance of the IWKS method is better than that of the other methods. In other words, our method provides an effective way to identify the critical nodes in a comprehensive transportation network.

Conclusion

To avoid the serious consequences of traffic disruption caused by natural or man-made disasters, a novel method is proposed to identify critical nodes based on the weighted k-shell model. The diversity of transportation modes, independent transportation ability, and connectivity of the node are considered as multiple attributes to determine the ranking of the nodes in the MTCN. To illustrate its effectiveness, the proposed method is compared with the existing methods based on the comprehensive transportation network in Zhejiang (China). The experimental results show that our method is reasonable at both the physical and transportation levels, and it has better performance in identifying critical nodes. The MCTN topology model can more accurately describe the coupling relationship of multiple modes of transportation. Also, the E-type and D-type nodes with a large degree, closeness centrality, and k-shell value should be given priority and key protection to improve their reliability against risks. Three attributes were used to calculate the weight in the proposed method, which is also feasible to select a few more attributes. The proposed method takes into account more comprehensive factors and can provide a more suitable reference for national or provincial managers to make decisions when a disaster strikes. This study may also provide a valuable reference for the application of the complex networks theory in the transportation system and improvement of transportation network reliability. However, it should be noted that this study only identified the critical nodes from the perspective of the transportation network topology. In the future, it is necessary to develop a method that combines actual data, such as transportation volume and economy.

CRediT authorship contribution statement

Longjian Wang: Conceptualization, Methodology, Formal analysis, Writing - original draft, Writing - review & editing, Visualization, Project administration. **Shaoya Zheng:** Investigation, Data curation, Writing - review & editing. **Yonggang Wang:** Writing - review & editing, Funding acquisition. **Longfei Wang:** Software, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Example explanation

This appendix provides the detailed calculation steps of the proposed method. To give an intuitive explanation of our algorithm, a simple network is constructed as an example. The simple network with 9 composite nodes is shown in Fig. 1. First, we need to calculate the edge type value using Eq. (1). Consider the coefficients of Eq. (1) are shown in Table A.1.

The edge type value of each edge is calculated and the results are shown in Table A.2. The edge type value of edge e_{h1} is taken as an example to calculate as the following.

$$c_{h1} = 3.1 \times [3 \times (1 \times 3.2 + 1 \times 2.2) + 1 \times 1 \times 3.2] = 60.14$$

Consider the operating time and distance of each edge are also known, as shown in Table A.3.

Then, the decision matrix A is obtained as follows:

$$A = \begin{vmatrix} 120 & 200 & 60.14 \\ 110 & 190 & 49.60 \\ 95 & 160 & 189.72 \\ 100 & 170 & 321.16 \\ 90 & 150 & 131.44 \\ 85 & 140 & 131.44 \\ 75 & 190 & 507.96 \\ 55 & 150 & 647.70 \\ 70 & 180 & 414.12 \\ 100 & 150 & 424.76 \\ 90 & 140 & 520.70 \\ 160 & 200 & 341.94 \\ 140 & 170 & 332.92 \\ 60 & 300 & 189.42 \\ 65 & 330 & 97.86 \\ 63 & 320 & 219.24 \\ 300 & 200 & 61.60 \\ 240 & 160 & 91.74 \end{vmatrix}$$

Next, matrix A is standardized using Eq. (3) or Eq. (4), and the calculation process of the element r_{11} is given as an example: $r_{11} = 55/120 = 0.4583$. The remaining elements of the matrix can be calculated in the same way, and the

standardized matrix R is as follows:

$$R = \begin{vmatrix} 0.4583 & 0.7000 & 0.0929 \\ 0.5000 & 0.7368 & 0.0766 \\ 0.5789 & 0.8750 & 0.2929 \\ 0.5500 & 0.8235 & 0.4958 \\ 0.6111 & 0.9333 & 0.2029 \\ 0.6471 & 1.0000 & 0.2029 \\ 0.7333 & 0.7368 & 0.7843 \\ 1.0000 & 0.9333 & 1.0000 \\ 0.7857 & 0.7778 & 0.6394 \\ 0.5500 & 0.9333 & 0.6558 \\ 0.6111 & 1.0000 & 0.8039 \\ 0.3438 & 0.7000 & 0.5279 \\ 0.3929 & 0.8235 & 0.5140 \\ 0.9167 & 0.4667 & 0.2925 \\ 0.8462 & 0.4242 & 0.1511 \\ 0.8730 & 0.4375 & 0.3385 \\ 0.1833 & 0.7000 & 0.0951 \\ 0.2292 & 0.8750 & 0.1416 \end{vmatrix}$$

Next, the matrix R is normalized using Eq. (5), and the calculation process of element h_{11} is given as an example: $h_{11} = 0.4583/10.8106 = 0.0424$. The remaining elements of the matrix can be calculated in the same way, and the normalized matrix H is as follows:

$$H = \begin{vmatrix} 0.0424 & 0.0504 & 0.0127 \\ 0.0462 & 0.0531 & 0.0105 \\ 0.0535 & 0.0631 & 0.0401 \\ 0.0509 & 0.0593 & 0.0679 \\ 0.0565 & 0.0673 & 0.0278 \\ 0.0599 & 0.0721 & 0.0278 \\ 0.0678 & 0.0531 & 0.1078 \\ 0.0925 & 0.0673 & 0.1368 \\ 0.0727 & 0.0560 & 0.0875 \\ 0.0509 & 0.0673 & 0.0897 \\ 0.0565 & 0.0721 & 0.1100 \\ 0.0318 & 0.0504 & 0.0722 \\ 0.0363 & 0.0593 & 0.0703 \\ 0.0848 & 0.0336 & 0.0400 \\ 0.0783 & 0.0306 & 0.0207 \\ 0.0808 & 0.0315 & 0.0463 \\ 0.0170 & 0.0504 & 0.0130 \\ 0.0212 & 0.0631 & 0.0194 \end{vmatrix}$$

Table A.1
The coefficients of Eq. (1) for the example.

	A-type	B-type	C-type	D-type	E-type
θ_i, θ_j	1	2	3	4	5
	Waterway	Aviation	Highway	Conventional railway	High-speed railway
σ_k	1.1	2.1	3.1	4.1	5.1
δ_i^k, δ_j^k	1.2	2.2	3.2	4.2	5.2

Table A.2
The edge type value of each edge.

Edge	Edge type value	Edge	Edge type value	Edge	Edge type value
e_{h1}	60.14	e_{s1}	507.96	e_{r4}	332.92
e_{h2}	49.6	e_{s2}	647.7	e_{f1}	189.42
e_{h3}	189.72	e_{s3}	414.12	e_{f2}	97.86
e_{h4}	321.16	e_{r1}	424.76	e_{f3}	219.24
e_{h5}	131.44	e_{r2}	520.7	e_{w1}	61.6
e_{h6}	131.44	e_{r3}	341.94	e_{w2}	91.74

Table A.3
The attribute values of each edge.

Edge	t (min)	d (km)	Edge	t (min)	d (km)	Edge	t (min)	d (km)
e_{h1}	120	200	e_{s1}	75	190	e_{r4}	140	170
e_{h2}	110	190	e_{s2}	55	150	e_{f1}	60	300
e_{h3}	95	160	e_{s3}	70	180	e_{f2}	65	330
e_{h4}	100	170	e_{r1}	100	150	e_{f3}	63	320
e_{h5}	90	150	e_{r2}	90	140	e_{w1}	300	200
e_{h6}	85	140	e_{r3}	160	200	e_{w2}	240	160

Then, we can calculate the information entropy value H_y using Eq. (6), and the following calculation process of H_1 is given as an example:

$$\begin{aligned} H_1 = -\frac{1}{\ln 18}(&0.0424 \times \ln 0.0424 + 0.0463 \times \ln 0.0463 + 0.0535 \times \ln 0.0535 + 2 \times 0.0509 \times \ln 0.0509 \\ &+ 2 \times 0.0565 \times \ln 0.0565 + 0.0599 \times \ln 0.0599 + 0.0678 \times \ln 0.0678 + 0.0925 \times \ln 0.0925 \\ &+ 0.0727 \times \ln 0.0727 + 0.0318 \times \ln 0.0318 + 0.0363 \times \ln 0.0363 + 0.0848 \times \ln 0.0848 \\ &+ 0.0783 \times \ln 0.0783 + 0.0808 \times \ln 0.0808 + 0.0170 \times \ln 0.0170 + 0.0212 \times \ln 0.0212) = 0.9741 \end{aligned}$$

In the same way, $H_2 = 0.9903$ and $H_3 = 0.9198$.

Next, the weight of the attributes can be calculated using Eq. (7), and the calculation process of weight w_1 is as follows:

$$w_1 = (1 - 0.9741) / [(1 - 0.9741) + (1 - 0.9903) + (1 - 0.9198)] = 0.2236$$

The remaining values can be calculated in the same way as above, $w_2 = 0.0838$ and $w_3 = 0.6926$.

Next, we can calculate the weight w_x of each edge using Eq. (8), and the following calculation process of weight w_{h1} is given as an example:

$$w_{h1} = 0.2236 \times 0.4583 + 0.0838 \times 0.7000 + 0.6926 \times 0.0929 = 0.2255$$

The remaining results are listed in Table A.4.

Next, the weight W_{ij} of each composite edge can be calculated using Eqs. (9) and (10), and the following calculation process of weight W_{56} is given as an example: $W_{56} = 0.5354 + 0.6554 = 1.1908$. The remaining results are shown in Table A.5.

Then, the values of degree k_i, closeness centrality C_c (i), and weight $\sum_j^{k_i} W_{ij}$ of each composite node are calculated and the results are shown in Table A.6.

Next, the value of the degree, closeness centrality, and weight of each composite node is standardized by Eq. (4), and the following calculation process of the degree of V_1 is given as an example: $k_1^* = 10 \times 3/6 = 5$. The remaining values can be calculated in the same way as above, and the results are listed in Table A.7.

Lastly, the node importance value of V_1 can be calculated by Eq. (15) as follows:

$$s_1^* = \sqrt[5]{5^2 \times 1.9834 \times 5.8140^2} = 4.4143$$

According to the above illustration, the IWKS value of each composite node can be obtained, and the results and ranking of the nodes are shown in Table A.8.

Table A.4
The weight of each edge.

Edge	Weight	Edge	Weight	Edge	Weight
e_{h1}	0.2255	e_{s1}	0.7689	e_{r4}	0.5129
e_{h2}	0.2266	e_{s2}	0.9944	e_{f1}	0.4467
e_{h3}	0.4056	e_{s3}	0.6837	e_{f2}	0.3294
e_{h4}	0.5354	e_{r1}	0.6554	e_{f3}	0.4663
e_{h5}	0.3554	e_{r2}	0.7772	e_{w1}	0.1655
e_{h6}	0.3690	e_{r3}	0.5012	e_{w2}	0.2226

Table A.5
The weight of each composite edge.

Edge	Weight	Edge	Weight	Edge	Weight	Edge	Weight
e_{12}	0.2255	e_{25}	0.4056	e_{47}	0.1655	e_{58}	1.7716
e_{15}	0.4467	e_{36}	0.7244	e_{56}	1.1908	e_{78}	0.7238
e_{17}	0.3294	e_{45}	0.7689	e_{57}	0.4663	e_{89}	1.1966
e_{23}	0.2266						

Table A.6
The values of degree, closeness centrality, and weight of composite nodes.

Node	Degree	Closeness	Weight	Node	Degree	Closeness	Weight
V_1	3	0.0025	1.0016	V_6	2	0.0031	1.9152
V_2	3	0.0033	0.8577	V_7	4	0.0028	1.6850
V_3	2	0.0026	0.9510	V_8	3	0.0035	3.6920
V_4	2	0.0028	0.9344	V_9	1	0.0023	1.1966
V_5	6	0.0043	5.0499				

Table A.7
The standardized values of degree, closeness centrality, and weight of composite nodes.

Node	Degree	Closeness	Weight	Node	Degree	Closeness	Weight
V_1	5.0000	5.8140	1.9834	V_6	3.3333	7.2093	3.7926
V_2	5.0000	7.6744	1.6984	V_7	6.6667	6.5116	3.3367
V_3	3.3333	6.0465	1.8832	V_8	5.0000	8.1395	7.3110
V_4	3.3333	6.5116	1.8503	V_9	1.6667	5.3488	2.3696
V_5	10.0000	10.0000	10.0000				

Table A.8
The values of IWKS and ranking of each composite node.

Node	IWKS	Ranking	Node	IWKS	Ranking
V_1	4.4143	6	V_6	4.6569	5
V_2	4.7820	4	V_7	5.7505	3
V_3	3.7734	8	V_8	6.5558	2
V_4	3.8732	7	V_9	2.8509	9
V_5	10.0000	1			

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